Research Journal

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September

Sep. 16, 2025 — Tuesday

- Revisited the credentials provided by the HPC facility at IUCAA last year – they are working fine
- Here are the current credentials:
  - **VPN**
    * Server: `moon.iucaa.in:4433`
    * Username: `parag.bhattacharya`
    * Password: `P@^@GBvpn2*2486`
    * Domain: `LocalDomain`
  - **ASSC account**
    * Command: `ssh -X paragb@192.168.11.34`
    * Password: *same as Win11*
  - **Using the installed software**

    ```
    source xraysoftinit
    heainit
    sas21init
    sas21
    ```

  - **Using Python3.10 etc.**

    ```
    module switch python/python310
    ```

  - **LDAP account credentials (at IUCAA)**
    * Username: `parag`
    * Password: `Saturn$890`

Sep. 17, 2025 — Wednesday

- Hopefully I will be able to do some polarimetric study of science data from IXPE, as proposed by Dr. Monmoyuri Baruah yesterday.
- The target for doing some work is 6 – 8 October 2025, during NEMA-XI at RGU, Arunachal Pradesh.
- I have two documents that I need to study:
  1. Bobrikova *et al.* (2024) – sent by Dr. Monmoyuri Baruah
  2. IXPE Quick Start Data Analysis Guide – downloaded from HEASARC
- To get started with the data analysis, I want to download the IXPE L2 data for two sources:
  1. GX 13+1 – a weakly magnetized neutron star (WMNS) studied by Bobrikova *et al.*
  2. Mrk 501 – a blazar which is provided as an example in the IXPE Quick Start Guide
- The science data are to be downloaded from the legacy HEASARC browse window as per the following parameters:
  - Object Name or Coordinates: `Mrk 501`
    Mission: `IXPE`
    Observation window: `July 2022`
  - Object Name or Coordinates: `GX 13+1`
    Mission: `IXPE`
    Observation window: `October 2023`

  Note: The current Xamin browse portal doesn't seem to load

Sep. 21, 2025 — Sunday

- Downloaded the L2 event files for the following dataset:
  1. `Mrk 501`
  2. `GX 13+1`
- Commands for file transfer using SCP – from Contabo Blog
- Commands for Mrk 501
  - Uploaded Mrk 501 `.tar` file from PC to ASSC server:

```
scp E:\Research\x-ray_polarimetry\data\raw\mrk-501.tar
 paragb@192.168.11.34:/home/paragb/xray-pol/dataset
```

- Navigated to data folder:

  ```
  cd /home/paragb/xray-pol/dataset
  ```

- Extracted data from the `.tar` file

  ```
  tar -xvf mrk-501.tar
  ```

- The extracted L2 files are contained in the folder:
  `/home/paragb/xray-pol/dataset/01004701/event_l2`
- The data folder has been accordingly renamed as:
  `mv -v 01004701/ mrk-501/`
- So the extracted L2 files are now contained in the folder:
  `/home/paragb/xray-pol/dataset/mrk-501/event_l2`
- The `.gz` files are unzipped as:
  `gunzip -v *.gz`

Sep. 22, 2025 — Monday

- Downloaded the relevant CALDB files for IXPE to the ASSC server using the commands

  ```
  export CALDB=/home/paragb/caldb/
  cd $CALDB
  wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/
              ixpe/gpd/goodfiles_ixpe_gpd_20250225.tar.gz
  wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/
              ixpe/xrt/goodfiles_ixpe_xrt_20241028.tar.gz
  gunzip -v goodfiles_*.gz
  tar -xvf goodfiles_*.tar
  rm -rv goodfiles_*.tar
  ```

- Downloaded the data `.tar` files for Mrk 501 directly to the ASSC server.

Sep. 23, 2025 — Tuesday

- Began download of Heasoft v6.35.2 from remote server using the following command after opening the terminal in the desired folder.

  ```
  wget -v https://heasarc.gsfc.nasa.gov/FTP/
      software/lheasoft/lheasoft6.35.2/heasoft-6.35.2src.tar.gz
  ```

- Installed MiKTeX using the following commands (MiKTeX webpage):

```
curl -fsSL https://miktex.org/download/key |
    sudo gpg --dearmor -o /usr/share/keyrings/miktex.gpg
echo "deb [signed-by=/usr/share/keyrings/miktex.gpg]
    https://miktex.org/download/ubuntu jammy universe" |
    sudo tee /etc/apt/sources.list.d/miktex.list
sudo apt-get update
sudo apt-get install miktex
miktexsetup finish
initexmf --set-config-value [MPM]AutoInstall=1
```

- Installed TeXstudio using the following commands:

```
sudo add-apt-repository ppa:sunderme/texstudio
sudo apt update
sudo apt install texstudio
```

- The `.tar` data files have been downloaded into the ASSC server at

```
/home/paragb/xray-pol/dataset/mrk-501/01004701/
```

- All `.gz` files have been extracted in the following folders:

```
auxil    event_l1    event_l2    hk
```

- To view the images on DS9: `ds9 ixpe01004701_det1_evt2_v01.fits &`
  - Enhancement of the image:
    `scale` → `log`
    `color` → `heat`
  - Creation of a source region:
    In the menu: e.g. `Region` → `Shape` → `Circle`
    `edit` → `region` → click + drag to draw the region
    Double-click on region to open the region dialog. Center in `fk5`, radius in `arcsec`.
  - Creation of a background region:
    In the menu: e.g. `Region` → `Shape` → `Annulus`
    `edit` → `region` → click + drag to draw the region
    Double-click on region to open the region dialog. Center in `fk5`, radius in `arcsec`.

- The region files are created in the L2 event folder. These files contain the following:
  - Inside `src.reg`:

    ```
    circle(16:53:51.766,+39:45:44.41,60.0")
    ```

  - Inside `bkg.reg`:

    ```
    circle(16:53:51.766,+39:45:44.41,252.0")
    -circle(16:53:51.766,+39:45:44.41,132.0")
    ```

- Extraction of spectra
  - Initializing `xselect`

    ```
    xselect prefix=mrk501
    xsel> read event "ixpe01004701_det1_evt2_v01.fits"
    ```

  - Extracting the source spectra (e.g. detector 1):

    ```
    xsel> filter region "src.reg"
    xsel> extract SPEC stokes=NEFF
    xsel> save spec mrk-501_d01_src_
    ```

    This creates 3 source spectra files, namely
    `mrk-501_d01_src_I.pha`
    `mrk-501_d01_src_Q.pha`
    `mrk-501_d01_src_U.pha`
  - Extracting the background spectra (e.g. detector 1):

    ```
    xsel> clear region
    xsel> filter region "bkg.reg"
    xsel> extract SPEC stokes=NEFF
    xsel> save spec mrk-501_d01_bkg_
    ```

    This creates 3 background spectra files, namely
    `mrk-501_d01_bkg_I.pha`
    `mrk-501_d01_bkg_Q.pha`
    `mrk-501_d01_bkg_U.pha`
  - Finding the appropriate RMF files:
    1. Find the observation in the `ixmaster` catalogue using the HEASARC search form
    2. Copy the entry for the object of interest under the `time` column, e.g. `2022-07-09 23:23:24.082` for the Mrk 501 obs. ID 01004701

3. Create a CALDB variable that points to the folder containing the IXPE calibration database as
   ```
   export CALDB=/home/paragb/caldb
   ```

4. Use the `time` value in the `quzcif` command, for detector unit 1 (DU1) of the gas pixel detector instrument (GPD), as follows:
   ```
   quzcif IXPE GPD DU1 - MATRIX 2022-07-09 23:23:24.082 -
   ```

   This would give the following output:
   ```
   /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
       ixpe_d1_20211209_01.rmf 1
   /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
       ixpe_d1_20211209_alpha075_01.rmf 1
   ```

5. We have to use the file whose name contains the string `alpha075` when the events are weighted by either the `NEFF` or `SIMPLE` weighting schemes. In our case, we used `NEFF` while extracting the source and background spectra and so we can create a soft link to the `alpha075` file as follows:
   ```
   ln -s /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
       ixpe_d1_20211209_alpha075_01.rmf mrk-501_d01_rmf.fits
   ```

   This creates a symbolic link file named `mrk-501_d01_rmf.fits` which can be used as the RMF file in XSPEC later.

- Creating the MRF (for Q and U spectra) and ARF (for I spectrum) files:

  1. Sent the following error report to NASA Helpdesk:
     ```
     Recently I have been introduced to IXPE science data analysis
     and I have been trying to replicate the results of the
     walkthrough in section 5 of the IXPE quick start analysis guide.

     I have been able to successfully generate the I, Q and U
     spectra as described in the aforementioned document.
     However, I am running into an error when I try to create
     the MRF file for the Q and U spectra.

     I have copied the attitude file from the /hk folder into
     the /event_l2 folder. The input and output of the console
     are as follows:
     [paragb@localhost event_l2]$ ixpecalcarf \
     > evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
     > attfiles=ixpe01004701_det1_att_v01.fits.gz \
     > arfout=mrk-501_d01.mrf \
     ```

```
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
        [ixpe01004701_det1_att_v01.fits]
        attfiles=ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'

The issue persists even when I provide the full path name
of the attitude file:
[paragb@localhost event_l2]$ ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
> attfiles=/home/paragb/xray-pol/dataset/mrk-501/
    01004701/event_l2/ixpe01004701_det1_att_v01.fits.gz \
> arfout=mrk-501_d01.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
    [attfiles=ixpe01004701_det1_att_v01.fits.gz]
        /home/paragb/xray-pol/dataset/mrk-501/01004701/event_l2/
        ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'

I am bamboozled because I have followed the exact same steps
as given in the document. I am unable to figure out as
to where I have been going wrong.

I would be grateful to receive advice from you in this matter.
```

Sep. 24, 2025 — Wednesday

- Received the following response from NASA Helpdesk:

```
Hello,

We were unable to reproduce the error you reported.
To help resolve this issue, please try the following:

1. Execute 'punlearn ixpecalcarf' to clean your parameter
file, as we suspect there may be a stale parameter
value causing the problem.

2. Please also check your attitude file configuration,
as your output shows the attitude file name appearing
inconsistently with and without the .gz extension.
```

```
Please let us know if this resolves the issue or if
you continue to experience problems.

Thanks,
Mihoko Yukita
```

- Installed Heasoft v6.35.2 on laptop.

Sep. 25, 2025 — Thursday

- Started analysis of data for Mrk 501 locally on my laptop.
- Installed Heasoft 6.35.2 locally on my PC (running on Ubuntu 22.04 LTS).
- Remote access for CALDB set up in the folder:
  `/home/pararover/Software/astro-soft/caldb`
  which contains the files `alias_config.fits` and `caldb.config`
- The `.bashrc` file is appended with the following lines:

```
CALDBCONFIG=/home/pararover/Software/astro-soft/caldb/
    caldb.config; export CALDBCONFIG
CALDBALIAS=/home/pararover/Software/astro-soft/caldb/
    alias_config.fits; export CALDBALIAS
CALDB=https://heasarc.gsfc.nasa.gov/FTP/caldb; export CALDB
```

- Data from HEASARC are downloaded and extracted into the following folder:

```
/home/pararover/Documents/Research/xray-pol/dataset/mrk-501/01004701
```

Sep. 28, 2025 — Sunday

- Downloaded `.tar` files for IXPE observations GX 13+1, with the Obs. ID. 02006801, in two batches because HEASARC does not allow downloads more than 2 GB.
  1. `event_l1/` for ~ 1.6 GB
  2. `event_l2/`, `auxil/` and `hk/` for ~ 700 MB
- The above two batches of `.tar` files need to be extracted and combined together into a single folder named `gx13+1/` with the following folder hierarchy:
  - `gx13+1/`
    * `02006801`

- auxil/
- event_l1/
- event_l2/
- hk/

October

Oct. 01, 2025 — Wednesday

- Found this good paper on polarized X-ray emission from a BH XRB named 4U 1630-47
  https://iopscience.iop.org/article/10.3847/2041-8213/acd77b
- For their analysis Rawat *et al.* have used IXPEOBSSIM
  https://ixpeobssim.readthedocs.io/en/latest/index.html
- I intend to explore IXPEOBSSIM further in this direction of research. To this end, I have installed the same on my PC using the command (after running `heainit`):
  ```
  pip3 install ixpeobssim
  ```
- IXPEOBSSIM requires the following Python packages to run:
  `numpy scipy matplotlib astropy regions skyfield`
- I have upgrade all of the above packages individually by running the command:
  ```
  pip3 install <<package-name>> --upgrade
  ```
- Before proceeding, I need to look into the tutorials provided here:
  https://ixpeobssim.readthedocs.io/en/latest/tutorial.html
- Logs from my tryout of the tutorial:

```
>>> import numpy
>>>
>>> from ixpeobssim.config import file_path_to_model_name
>>> About to create folder /home/pararover/ixpeobssimdata...
>>> Folder succesfully created.
>>> About to create folder /home/pararover/ixpeobssimdata/benchmarks...
>>> Folder succesfully created.
>>> About to create folder /home/pararover/ixpeobssimauxfiles...
>>> Folder succesfully created.
Matplotlib is building the font cache; this may take a moment.
/home/pararover/.local/lib/python3.10/site-packages/matplotlib/
    projections/__init__.py:63:
    UserWarning: Unable to import Axes3D. This may be due to
    multiple versions of Matplotlib being installed
```

```
    (e.g. as a system package and as a pip package). As a result,
    the 3D projection is not available.
  warnings.warn("Unable to import Axes3D.
    This may be due to multiple versions of "
>>> from ixpeobssim.srcmodel.roi import xPointSource, xROIModel
>>> from ixpeobssim.srcmodel.spectrum import power_law
>>> from ixpeobssim.srcmodel.polarization import constant
>>>
```

Oct. 16, 2025 — Thursday

- The solution provided by the IXPE Helpdesk is working.
- Was able to run `ixpecalcarf` to generate MRF files for D01, D02 and D03.
- However, the following warning appeared:

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
/home/pararover/.local/lib/python3.10/site-packages/matplotlib/
projections/__init__.py:63: UserWarning: Unable to import Axes3D.
This may be due to multiple versions of Matplotlib being
installed (e.g. as a system package and as a pip package). As a
result, the 3D projection is not available.
  warnings.warn("Unable to import Axes3D. This may be due to
  multiple versions of "
```

- The `ixpecalarf` task has been run as follows:

  1. For detector #1:
     To generate an MRF (for Q and U spectra):

     ```
     pararover@shadowfax: ixpecalcarf \
     > evtfile=ixpe01004701_det1_evt2_v01.fits \
     > attfiles=ixpe01004701_det1_att_v01.fits \
     > arfout=mrk-501_d01_QU.mrf \
     > specfile=none radius=1.0 weight=1 resptype=mrf
     ```

     To generate an ARF (for I spectrum):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits \
> attfiles=ixpe01004701_det1_att_v01.fits \
> arfout=mrk-501_d01_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf
```

2. For detector #2:

   To generate an MRF (for Q and U spectra):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
```

   To generate an ARF (for I spectrum):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf
```

3. For detector #3:

   To generate an MRF (for Q and U spectra):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
```

   To generate an ARF (for I spectrum):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf
```

- The list of necessary files for detector #1 are:

```
mrk-501_d01_src_I.pha
```

```
mrk-501_d01_src_Q.pha
mrk-501_d01_src_U.pha
mrk-501_d01_bkg_I.pha
mrk-501_d01_bkg_Q.pha
mrk-501_d01_bkg_U.pha
mrk-501_d01.rmf -> ixpe_d1_20211209_alpha075_01.rmf
mrk-501_d01_I.arf
mrk-501_d01_QU.mrf
```

- The list of necessary files for detector #2 are:

```
mrk-501_d02_src_I.pha
mrk-501_d02_src_Q.pha
mrk-501_d02_src_U.pha
mrk-501_d02_bkg_I.pha
mrk-501_d02_bkg_Q.pha
mrk-501_d02_bkg_U.pha
mrk-501_d02.rmf -> ixpe_d2_20211209_alpha075_01.rmf
mrk-501_d02_I.arf
mrk-501_d02_QU.mrf
```

- The list of necessary files for detector #3 are:

```
mrk-501_d03_src_I.pha
mrk-501_d03_src_Q.pha
mrk-501_d03_src_U.pha
mrk-501_d03_bkg_I.pha
mrk-501_d03_bkg_Q.pha
mrk-501_d03_bkg_U.pha
mrk-501_d03.rmf -> ixpe_d2_20211209_alpha075_01.rmf
mrk-501_d03_I.arf
mrk-501_d03_QU.mrf
```

- Commands for spectrum file grouping of detector #1:

  1. For I spectrum:

     ```
     fthedit mrk-501_d01_src_I.pha keyword=BACKFILE operation=add \
     value=" 'mrk-501_d01_bkg_I.pha'" longstring=YES

     fthedit mrk-501_d01_src_I.pha keyword=RESPFILE operation=add \
     value=" 'mrk-501_d01.rmf'" longstring=YES
     ```

```
fthedit mrk-501_d01_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d01_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_Q.pha'" longstring=YES

fthedit mrk-501_d01_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmf'" longstring=YES

fthedit mrk-501_d01_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.mrf'" longstring=YES
```

3. For U spectrum:

```
fthedit mrk-501_d01_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_U.pha'" longstring=YES

fthedit mrk-501_d01_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmf'" longstring=YES

fthedit mrk-501_d01_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.mrf'" longstring=YES
```

- Commands for spectrum file grouping of detector #2:

1. For I spectrum:

```
fthedit mrk-501_d02_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_I.pha'" longstring=YES

fthedit mrk-501_d02_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmf'" longstring=YES

fthedit mrk-501_d02_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d02_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmf'" longstring=YES

fthedit mrk-501_d02_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

3. For U spectrum:

```
fthedit mrk-501_d02_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_U.pha'" longstring=YES

fthedit mrk-501_d02_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmf'" longstring=YES

fthedit mrk-501_d02_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

- Commands for spectrum file grouping of detector #3:

1. For I spectrum:

```
fthedit mrk-501_d03_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_I.pha'" longstring=YES

fthedit mrk-501_d03_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmf'" longstring=YES

fthedit mrk-501_d03_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d03_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_Q.pha'" longstring=YES

fthedit mrk-501_d03_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmf'" longstring=YES

fthedit mrk-501_d03_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_QU.mrf'" longstring=YES
```

3. For U spectrum:

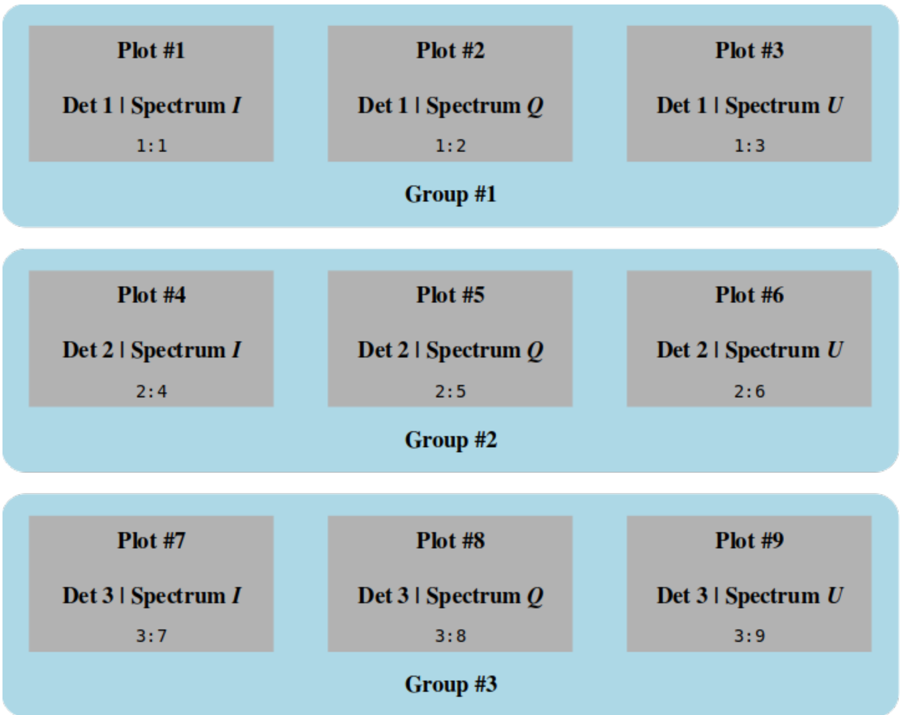
```
fthedit mrk-501_d03_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_U.pha'" longstring=YES

fthedit mrk-501_d03_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmf'" longstring=YES

fthedit mrk-501_d03_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_QU.mrf'" longstring=YES
```

Oct. 17, 2025 — Friday

- Spectral fitting of all data groups to be performed as per the Quick Start Manual.
- Understanding the data and plot groups in XSPEC:



- The above data groups are created in XSPEC as follows:

```
data 1:1 mrk-501_d01_src_I.pha
data 1:2 mrk-501_d01_src_Q.pha
data 1:3 mrk-501_d01_src_U.pha

data 2:4 mrk-501_d02_src_I.pha
```

```
data 2:5 mrk-501_d02_src_Q.pha
data 2:6 mrk-501_d02_src_U.pha

data 3:7 mrk-501_d03_src_I.pha
data 3:8 mrk-501_d03_src_Q.pha
data 3:9 mrk-501_d03_src_U.pha
```

Oct. 18, 2025 — Saturday

- Steps that were used to analyze the spectra of Mrk 501 in XSPEC are given below:

  I. Reading the data:

  ```
  XSPEC12> data 1:1 mrk-501_d01_src_I.pha
  XSPEC12> data 1:2 mrk-501_d01_src_Q.pha
  XSPEC12> data 1:3 mrk-501_d01_src_U.pha
  XSPEC12> data 2:4 mrk-501_d02_src_I.pha
  XSPEC12> data 2:5 mrk-501_d02_src_Q.pha
  XSPEC12> data 2:6 mrk-501_d02_src_U.pha
  XSPEC12> data 3:7 mrk-501_d03_src_I.pha
  XSPEC12> data 3:8 mrk-501_d03_src_Q.pha
  XSPEC12> data 3:9 mrk-501_d03_src_U.pha
  ```

  II. Filtering the photons within an energy bandwidth:

  ```
  XSPEC12> ignore *:**-2.0 8.0-**
  ```

  Here the first `*` means to apply this filter on all data groups. Then `**-2.0 8.0-**` means to ignore all photon counts for $E < 2\,\mathrm{keV}$ and $E > 8\,\mathrm{keV}$, i.e. only the the $2\,\mathrm{keV} \leqslant E \leqslant 8\,\mathrm{keV}$ bandwidth is to be considered.

  III. Applying the composite model:

  ```
  XSPEC12> model constant*tbabs*(polconst*powerlaw)
  ```

  Here `constant` models an energy-independent multiplicative factor, `tbabs` models the intervening absorption, `polconst` models a constant polarization at all energies and `powerlaw` models a simple photon power law.

  IV. Parameter #1 is frozen at 1. Parameters 7 and 13 are untied and thawed. The parameter list prior to fitting looks as follows:

  ```
  XSPEC12> show par

  Parameters defined:
  ```

```
========================================================================
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component  Parameter  Unit     Value
par  comp
Data group: 1
1     1   constant   factor              1.00000      frozen
2     2   TBabs      nH         10^22    1.00000      +/-  0.0
3     3   polconst   A                   1.00000      +/-  0.0
4     3   polconst   psi        deg      45.0000      +/-  0.0
5     4   powerlaw   PhoIndex            1.00000      +/-  0.0
6     4   powerlaw   norm                1.00000      +/-  0.0
Data group: 2
7     1   constant   factor              1.00000      +/-  0.0
8     2   TBabs      nH         10^22    1.00000      = p2
9     3   polconst   A                   1.00000      = p3
10    3   polconst   psi        deg      45.0000      = p4
11    4   powerlaw   PhoIndex            1.00000      = p5
12    4   powerlaw   norm                1.00000      = p6
Data group: 3
13    1   constant   factor              1.00000      +/-  0.0
14    2   TBabs      nH         10^22    1.00000      = p2
15    3   polconst   A                   1.00000      = p3
16    3   polconst   psi        deg      45.0000      = p4
17    4   powerlaw   PhoIndex            1.00000      = p5
18    4   powerlaw   norm                1.00000      = p6
________________________________________________________________________
```

V. Renormalizing and fitting the model to the spectrum gives the following parameter values:

```
========================================================================
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component  Parameter  Unit     Value
par  comp
Data group: 1
1     1   constant   factor              1.00000      frozen
2     2   TBabs      nH         10^22  0.541250     +/-  0.117460
3     3   polconst   A                 7.25453E-02  +/-  1.90183E-02
4     3   polconst   psi        deg    -49.1296     +/-  7.50921
5     4   powerlaw   PhoIndex          2.60742      +/-  5.68093E-02
6     4   powerlaw   norm              9.63365E-02  +/-  8.13714E-03
Data group: 2
```

```
7     1   constant   factor           1.00375      +/-  9.94110E-03
8     2   TBabs      nH        10^22  0.541250     = p2
9     3   polconst   A                7.25453E-02  = p3
10     3   polconst   psi      deg    -49.1296     = p4
11     4   powerlaw   PhoIndex        2.60742      = p5
12     4   powerlaw   norm            9.62997E-02  = p6
Data group: 3
13     1   constant   factor          0.924722     +/-  1.00195E-02
14     2   TBabs      nH       10^22  0.541250     = p2
15     3   polconst   A               7.25453E-02  = p3
16     3   polconst   psi      deg    -49.1296     = p4
17     4   powerlaw   PhoIndex        2.60742      = p5
18     4   powerlaw   norm            9.63365E-02  = p6
_________________________________________________________________


Fit statistic: Chi-Squared 131.57 using 149 bins,
          spectrum 1, group 1.
Chi-Squared      159.05     using 149 bins, spectrum 2, group 1.
Chi-Squared      122.69     using 149 bins, spectrum 3, group 1.
Chi-Squared      130.71     using 149 bins, spectrum 4, group 2.
Chi-Squared      118.58     using 149 bins, spectrum 5, group 2.
Chi-Squared      157.11     using 149 bins, spectrum 6, group 2.
Chi-Squared       60.87     using 149 bins, spectrum 7, group 3.
Chi-Squared       82.38     using 149 bins, spectrum 8, group 3.
Chi-Squared       69.58     using 149 bins, spectrum 9, group 3.
Total fit statistic                1032.53     with 1334 d.o.f.

Test statistic : Chi-Squared       1032.53     using 1341 bins.
Null hypothesis probability of 1.00e+00 with 1334 degrees of freedom
```

VI. We are interested in finding the errors in the polarization degree A and polarization angle ψ, which are represented by the parameters A and `psi` in the `polconst` model component.

- To find the 99% confidence intervals for, say data group 1, we have

```
XSPEC12> error 6.635 3
XSPEC12> error 6.635 4
```

- To find the 1σ confidence intervals for, say data group 1, we have

```
XSPEC12> error 1.0 3
XSPEC12> error 1.0 4
```

Note: We cannot compute the errors in the same parameters for the other two data groups (i.e. 2 and 3) because these parameters are tied to their counterparts in data group 1. Any attempt to do so will be flagged as an error by XSPEC as follows:

```
XSPEC12> error 6.635 9
*** Parameter 9 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
XSPEC12> error 6.635 10
*** Parameter 10 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
XSPEC12> error 6.635 15
*** Parameter 15 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
XSPEC12> error 6.635 16
*** Parameter 16 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
```

VII. To visualize the contour plot, we need the following commands:

```
XSPEC12> steppar 3 0.00 0.21 50 4 -90 0 60
XSPEC12> plot contour ,,4 1.386, 4.61 9.21 13.81
```

The `steppar` command generates a 2D grid of A and ψ values with 50 steps in the range $0 < A < 0.21$ and 60 steps in the range $-90^\circ < \psi < 0^\circ$.
The `plot contour` command plots contours at 50% (`1.386`), 90% (`4.61`), 99% (`9.21`) and 99.9% (`13.81`) confidence levels for 2D errors in χ^2.

- To estimate the MDP$_{99}$ using PIMMS, we need to do the following:

1. Determine the model flux in the $2\,\mathrm{keV} \leqslant E \leqslant 8\,\mathrm{keV}$ bandwidth:

```
XSPEC12> flux 2.0 8.0
Spectrum Number: 1, 2, 3
Data Group Number: 1
Model Flux  0.015886 photons (8.7966e-11 ergs/cm^2/s) range
 (2.0000 - 8.0000 keV)

Spectrum Number: 4, 5, 6
Data Group Number: 2
Model Flux  0.015945 photons (8.8297e-11 ergs/cm^2/s) range
```

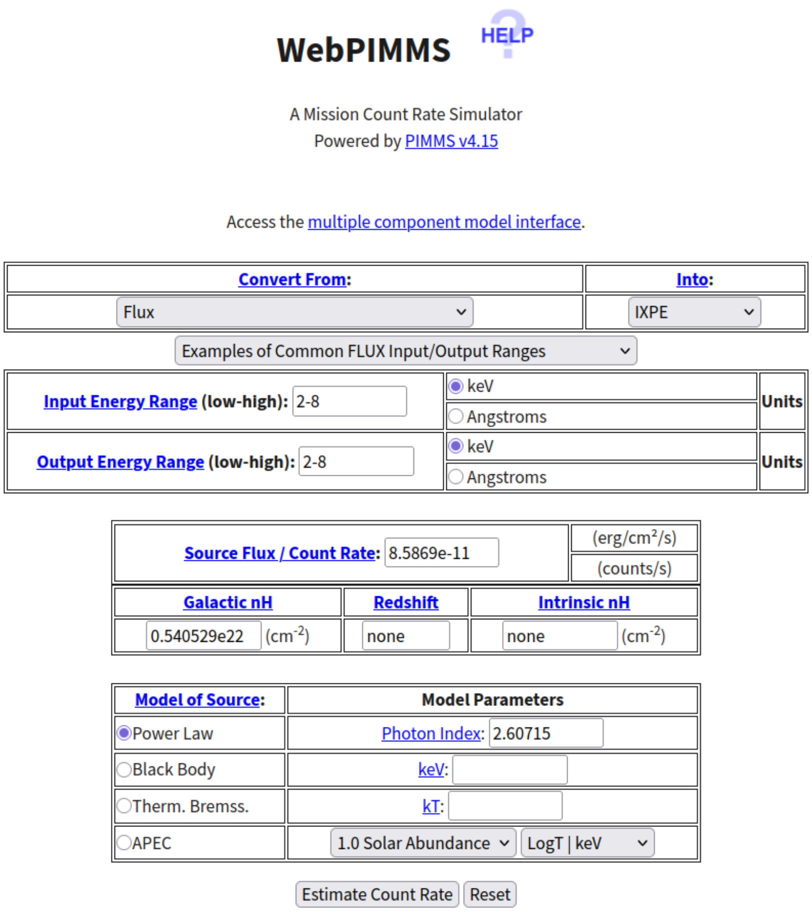
```
 (2.0000 - 8.0000 keV)

Spectrum Number: 7, 8, 9
Data Group Number: 3
Model Flux   0.01469 photons (8.1345e-11 ergs/cm^2/s) range
 (2.0000 - 8.0000 keV)
```

2. Collect the following values from the best-fit parameters so far:

$$n_H = 0.540\,529 \times 10^{22}\,\mathrm{cm}^{-2} \quad \text{(Galactic H column density)}$$
$$\Gamma = 2.60715 \quad \text{(photon index from powerlaw)}$$
$$\langle F\rangle = \left(\frac{8.7966 + 8.8297 + 8.1345}{3}\right) 10^{-11} = 8.5869 \times 10^{-11} \quad \text{(avg. flux from 3 detectors)}$$

3. Enter these values, along with some more, in WebPIMMS as shown:



4. WebPIMMS computes and displays the predicted count rates and MDP_{99} values using different algorithms, some of which are shown below: We find,

---

IXPE OPEN Modulation Factor

Modulation Factor = **0.3140**
In 10000.0 s, MDP_99 = **16.13%**
In 100000.0 s, MDP_99 = **5.10%**

IXPE GRAY TR21D

The Total count rate in the selected band:
PIMMS predicts **1.817E-01** cps with IXPE GRAY TR21D

IXPE GRAY ER21D

The Effective count rate in the selected band:
PIMMS predicts **1.549E-01** cps with IXPE GRAY ER21D

IXPE GRAY MR21D

The Modulated count rate in the selected band:
PIMMS predicts **6.300E-02** cps with IXPE GRAY MR21D

IXPE GRAY Modulation Factor

Modulation Factor = **0.4069**
In 10000.0 s, MDP_99 = **26.80%**
In 100000.0 s, MDP_99 = **8.48%**

for instance, under 'IXPE OPEN Modulation Factor', that for an exposure time of $100\,\mathrm{ks}$, the MDP_{99} is 5.15%.

$$T = 100\,\mathrm{ks} \implies \mathrm{MDP}_{99} = 5.10\% = 0.051$$

Now MDP_{99} scales with exposure time T as $\mathrm{MDP}_{99} \propto \dfrac{1}{\sqrt{T}}$. So we have

$$\frac{\mathrm{MDP}_{99}(\text{actual})}{\mathrm{MDP}_{99}(\text{simulated})} = \sqrt{\frac{T(\text{simulated})}{T(\text{actual})}}$$

Running the `show data` command gives us the actual exposure time (e.g. for I spectrum by detector #1) to be $97800\,\mathrm{s} = 97\,\mathrm{ks}$.
Therefore, substituting $T(\text{simulated}) = 100\,\mathrm{ks}$, $T(\text{actual}) = 97\,\mathrm{ks}$ and $\mathrm{MDP}_{99}(\text{simulated}) = 5.10\%$ above, we obtain

$$\mathrm{MDP}_{99}(\text{actual}) = \sqrt{\frac{100}{97}}(5.10\%) = 5.18\% = 0.0518$$

This implies that, for an *unpolarized source* with the same physical parameters, an IXPE observation will return a value $A > 0.0518$ only 1% (=100%-99%, the 99 is from MDP$_{99}$) of the time.

- The chain of logic is as follows:
    1. Scale PIMMS number because the real observation time was a bit shorter, i.e. MDP$_{99} = 5.18\%$.
    2. MDP$_{99}$ means "only 1% chance a zero-polarisation source would look this big or bigger."
    3. Measured value of 7.24% (from best-fit) is above 5.18%, and the measurement's uncertainty of 1.9% is small (i.e. $5.18/1.9 \approx 3.8$).
    4. Therefore the measured polarisation is very unlikely to be a random fluke – it is a highly probable real detection.